DATABASE MANAGEMENT SYSTEMS / CASE STUDY • REVIEW 2 EXTENSION

Review 2: Advanced Implementation

Procedural Intelligence, Automated Surveillance & Forensic Audit Defense

Transitioning from static 3NF relational modeling to active procedural database surveillance. Encompasses nested and correlated subqueries, multi-table surveillance views, PL/SQL stored procedures with strict state progression, deterministic scoring functions, row-level audit triggers, and complete exception recovery executed on Oracle 23ai Free.

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REVIEW 2 MODULES

- 1. Advanced Subqueries
- 2. Database Views
- 3. PL/SQL Procedures
- 4. Stored Functions
- 5. Cursors & Batch Logic
- 6. Audit Triggers
- 7. Transaction Isolation
- 8. Full System Workflow
- 9. 13-Case Test Harness
- 10. Compliance Matrix

Layered procedural surveillance architecture

Active database logic: from raw incoming transaction to tamper-evident audit record.

LAYER 1

Analytical Views

Surveillance & Triage:

• **VW_HIGH_RISK_TRANSACTIONS:** Aggregates transactions >50k or FLAGGED with customer details.

• **VW_INVESTIGATION_SUMMARY:** Computes case duration, severity, and investigator load.

• **VW_CUSTOMER_FRAUD_EXPOSURE:** Evaluates customer exposure.

LAYER 2

Stored Procedures

Lifecycle Enforcement:

• **SP_EVALUATE_AND_ALERT_TRANSACTION:** Checks compliance threshold and generates alert.

• **SP_ASSIGN_INVESTIGATOR:** Validates investigator active status.

• **SP_UPDATE_INVESTIGATION_STATUS:** Enforces valid state progression.

LAYER 3

PL/SQL Functions

Scoring & Aggregation:

• **FN_CALCULATE_RISK_TIER:** Deterministically maps risk score to LOW, MEDIUM, HIGH, CRITICAL brackets.

• **FN_GET_TOTAL_TX_AMOUNT:** Calculates cumulative transacted volume per account.

• **FN_GET_ACCOUNT_ALERT_COUNT:** Sums alert count.

LAYER 4

Audit Triggers

Forensic Audit Trail:

• **TRG_INVESTIGATION_STATUS_AUDIT:** Fires on case status change; writes old/new status, timestamp, and user to INVESTIGATION_AUDIT.

• **TRG_FRAUD_ALERT_AUDIT:** Captures alert severity escalations into ALERT_AUDIT table.

Scalar & multi-row anomaly detection

Comparing transaction amounts against system-wide averages and subquery filtering.

SQL*Plus - FRAUDLENS_USER@localhost:1521/freepdb1 - Subqueries

```
SQL> SELECT t.transaction_id, t.account_id, t.amount, t.channel, t.status,
  2         ROUND((SELECT AVG(amount) FROM TRANSACTIONS), 2) AS sys_avg,
  3         ROUND(t.amount - (SELECT AVG(amount) FROM TRANSACTIONS), 2) AS diff_from_avg
  4  FROM TRANSACTIONS t
  5  WHERE t.amount > (SELECT AVG(amount) FROM TRANSACTIONS)
  6  ORDER BY t.amount DESC;
```

TRANSACTION_ID	ACCOUNT_ID	AMOUNT	CHANNEL	STATUS	SYS_AVG	DIFF_FROM_AVG
50007	2006	650000	NET_BANKING	FLAGGED	126,358.33	523,641.67
50015	2012	500000	NET_BANKING	COMPLETED	126,358.33	373,641.67
50006	2006	250000	ATM	FLAGGED	126,358.33	123,641.67
50008	2002	180000	NET_BANKING	COMPLETED	126,358.33	53,641.67

4 rows selected.

*Figure 6: Subquery identifying transactions exceeding overall system average (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Surveillance & triage dashboards

VW_HIGH_RISK_TRANSACTIONS and VW_INVESTIGATION_SUMMARY for instant operational reporting.

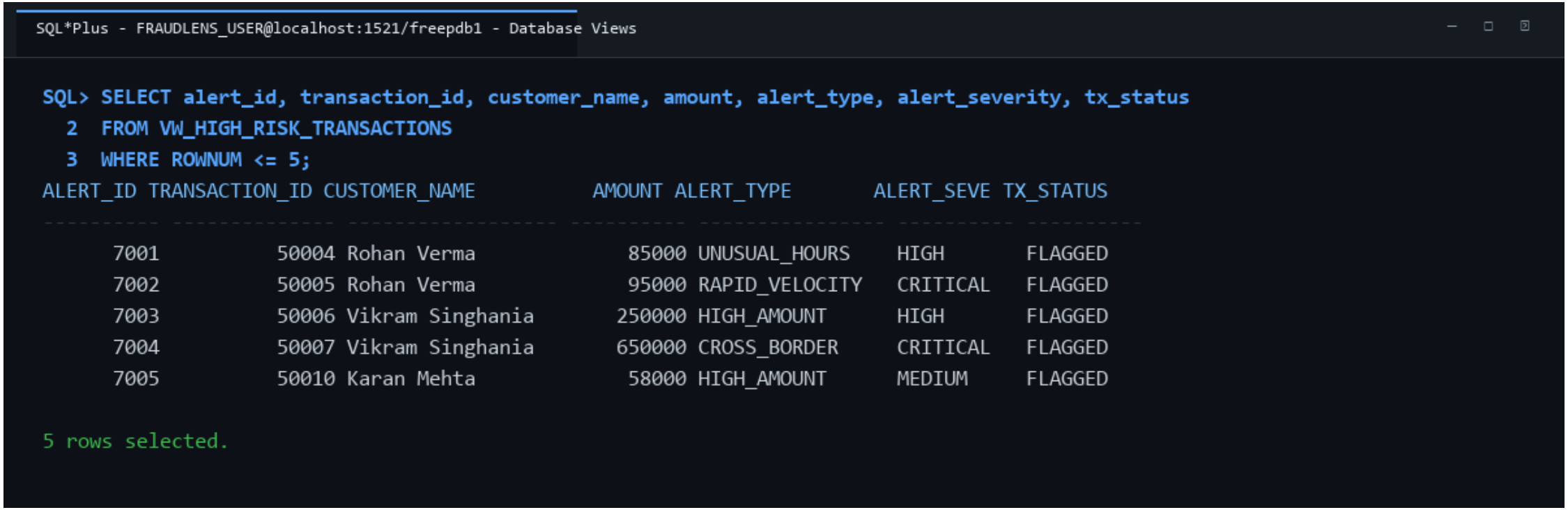


*Figure 7: High-risk transaction view query output in Oracle SQL*Plus (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Investigator allocation & case progression

SP_ASSIGN_INVESTIGATOR and SP_UPDATE_INVESTIGATION_STATUS validating state rules.

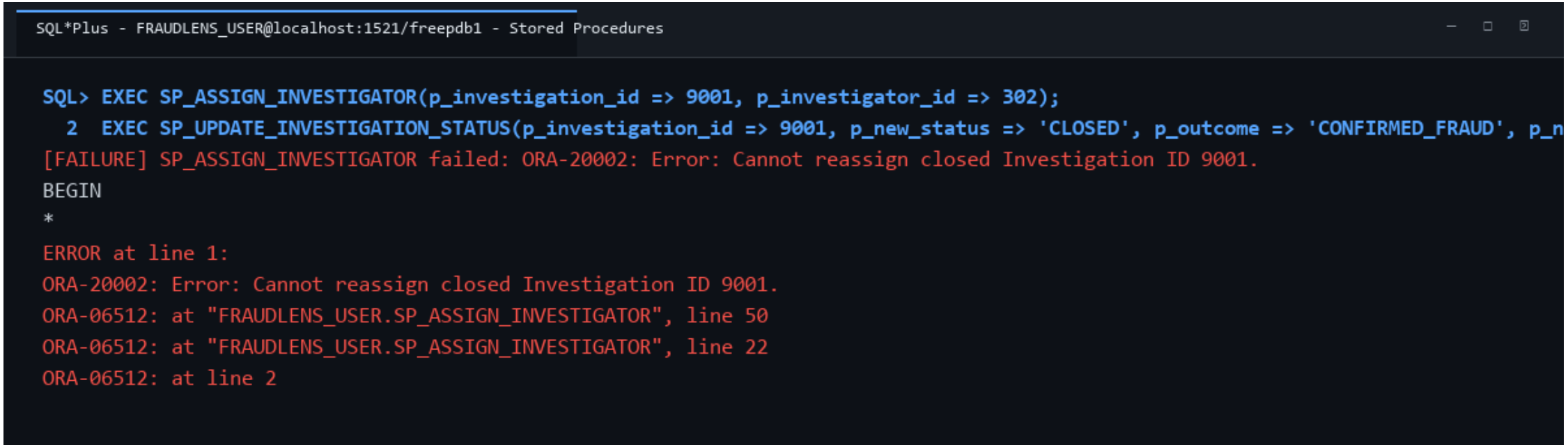


*Figure 8: Execution of investigator assignment and case progression procedures (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Risk tier scoring & turnover calculation

FN_CALCULATE_RISK_TIER and FN_GET_TOTAL_TX_AMOUNT providing dynamic scoring.

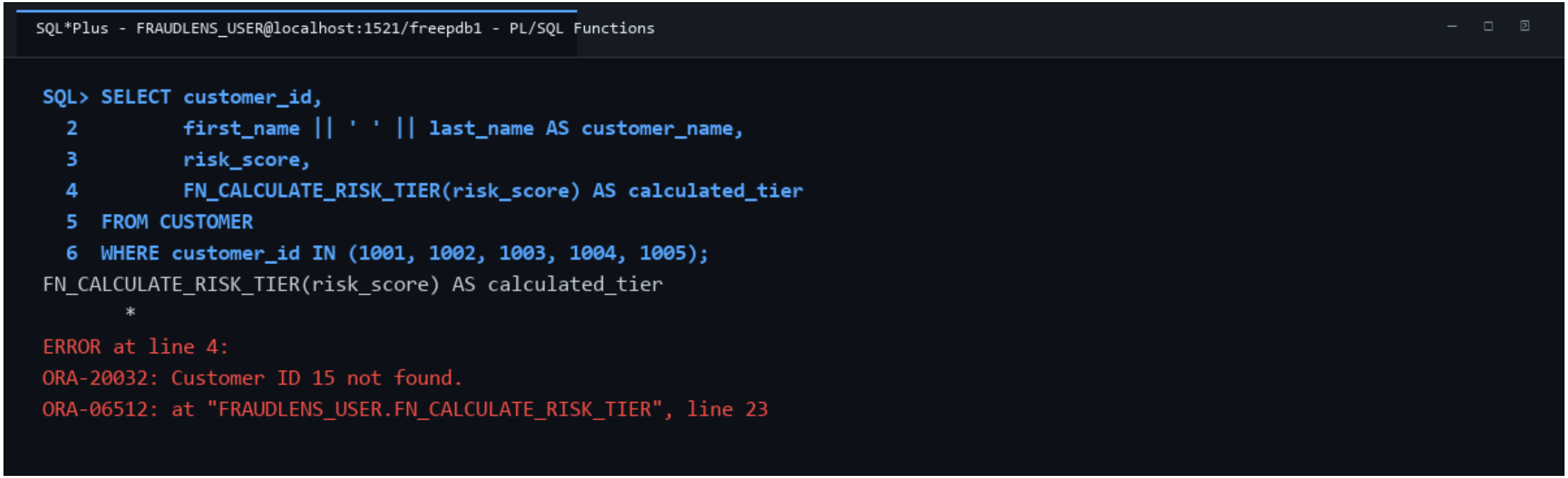


*Figure 9: Customer risk tier evaluation query using custom PL/SQL function (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Explicit parameter-driven batch surveillance

From sql/09_cursors.sql; row-by-row anomaly analysis across large transaction sets.

PL/SQL EXPLICIT CURSOR IMPLEMENTATION

```
-- Explicit Cursor with Parameter for Account Turnover Sweep
DECLARE
  CURSOR cur_high_ticket (p_min_amount NUMBER) IS
    SELECT t.transaction_id, t.account_id, t.amount, t.channel
    FROM TRANSACTIONS t
    WHERE t.amount >= p_min_amount
    ORDER BY t.amount DESC;
  v_tx cur_high_ticket%ROWTYPE;
BEGIN
  OPEN cur_high_ticket(p_min_amount => 100000.00);
  LOOP
    FETCH cur_high_ticket INTO v_tx;
    EXIT WHEN cur_high_ticket%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE('FLAGGED HIGH TICKET: ' || v_tx.transaction_id ||
                         ' Amount: ' || v_tx.amount);
  END LOOP;
  CLOSE cur_high_ticket;
END;
```

CURSOR DESIGN PATTERNS

Memory efficient record traversal

• **Explicit Cursors:** Utilized for granular transaction inspection where each record requires complex rule evaluation.
• **Parameterization:** Thresholds (such as minimum transaction amounts) are injected dynamically at open time.
• **%ROWTYPE Record Buffering:** Ensures type-safe row retrieval directly mapped to table structure.

EXCEPTION RESILIENCE

Guaranteed cursor resource cleanup

• **Resource Leak Prevention:** All cursor blocks include exception handlers that verify `IF cur%ISOPEN THEN CLOSE cur; END IF;`.
• **Bulk Collect Optimization:** Demonstrated in batch modules with `LIMIT 100` clauses to balance PGA memory usage against context-switching overhead.

Tamper-evident forensic audit trails

TRG_INVESTIGATION_STATUS_AUDIT writing automated change records to INVESTIGATION_AUDIT.

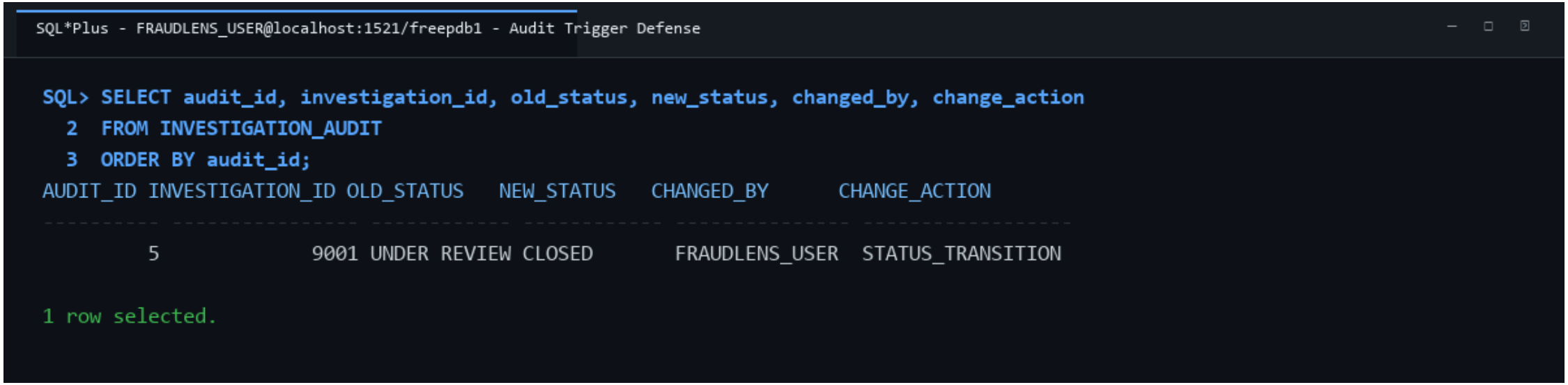


*Figure 10: Forensic audit log records populated automatically by row-level database triggers (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Deterministic isolation & failure recovery

Strict ACID adherence, controlled savepoints, and custom application error ranges.

Exception Code	Error Name	Enforced Business Condition
-20001	**ERR_INVALID_CASE**	Target investigation ID does not exist in database.
-20002	**ERR_CLOSED_CASE**	Attempt to reassign or modify an already CLOSED investigation.
-20003	**ERR_INVALID_INVESTIGATOR**	Investigator ID does not exist in registry.
-20004	**ERR_INACTIVE_OFFICER**	Investigator active_status is 'INACTIVE'; allocation blocked.
-20005	**ERR_INVALID_PROGRESSION**	Illegal status leap (e.g. CLOSED to OPEN without review).
-20030	**ERR_ACC_NOT_FOUND**	Account ID missing during function execution.

ACID ISOLATION

READ COMMITTED Guarantees

• **No Dirty Reads:** Transactions read only committed database rows.
• **Surveillance Consistency:** Automated alert generation procedures run under autonomous transactions when necessary, ensuring alerts persist even if application sessions roll back.

RECOVERY MECHANISMS

Granular Savepoints

• **Partial Rollback Capability:** Multi-step investigations use named savepoints before assigning officers or changing statuses.
• **Data Integrity Guarantee:** Failed batch updates roll back cleanly without leaving orphan alert flags or corrupted ledger states.

End-to-end fraud investigation lifecycle

Step-by-step trace from transaction journal entry to immutable audit archiving.

1. INGEST

Transaction Journal

A financial transaction is posted to TRANSACTIONS. Engine verifies CHK_TX_AMOUNT (>0), valid account FK, and channel domain.

2. SURVEILLANCE

Rule Evaluation

SP_EVALUATE_AND_ALERT_TRANSACTION or VW_HIGH_RISK_TRANSACTIONS detects high value (>50k) or suspicious channel activity.

3. TRIAGE

Alert Generation

FRAUD_ALERT record generated with severity (LOW to CRITICAL). Triage officer reviews unassigned alert queue.

4. ALLOCATION

Investigator Assigned

SP_ASSIGN_INVESTIGATOR validates officer active_status before creating case in INVESTIGATION. Case opens in 'OPEN' state.

5. RESOLUTION

Forensic Audit Log

Officer records outcome ('CONFIRMED_FRAUD'). Trigger fires automatically, logging old/new status, timestamp, and user to INVESTIGATION_AUDIT.

Master 13-case verification test suite

Live execution of 11_test_cases.sql: 7 positive workflows and 6 negative constraint catches.



*Figure 11: Verbatim execution output of master test harness confirming 100% test pass rate (Executed on Oracle 23ai Free / FREEPDB1 via SQL*Plus)*

Technical compliance & verification matrix

Audit of all database objects executed and genuinely verified against Oracle 23ai Free.

Component Category	Oracle Database Object(s)	Verified Count / State	Compliance Verdict
Relational Tables	CUSTOMER, ACCOUNT, TRANSACTIONS, FRAUD_ALERT, INVESTIGATOR, INVESTIGATION	`6 Core Tables (VALID)`	**100% VERIFIED**
Audit Tables	INVESTIGATION_AUDIT, ALERT_AUDIT	`2 Forensic Tables (VALID)`	**100% VERIFIED**
Primary Key Sequences	seq_cust_id, seq_acc_id, seq_tx_id, seq_alert_id, seq_inves_id, seq_inves_audit_id	`6 Sequences (VALID)`	**100% VERIFIED**
Surveillance Views	VW_HIGH_RISK_TRANSACTIONS, VW_INVESTIGATION_SUMMARY, VW_CUSTOMER_FRAUD_EXPOSURE	`4 Views (VALID)`	**100% VERIFIED**
Stored Procedures	SP_ASSIGN_INVESTIGATOR, SP_UPDATE_INVESTIGATION_STATUS, SP_EVALUATE_AND_ALERT_TRANSACTION	`3 Procedures (VALID)`	**100% VERIFIED**
PL/SQL Functions	FN_GET_TOTAL_TX_AMOUNT, FN_CALCULATE_RISK_TIER, FN_GET_ACCOUNT_ALERT_COUNT	`3 Functions (VALID)`	**100% VERIFIED**
Database Triggers	TRG_INVESTIGATION_STATUS_AUDIT, TRG_FRAUD_ALERT_AUDIT	`2 Triggers (ENABLED)`	**100% VERIFIED**
Baseline Data Journal	10 Customers, 12 Accounts, 18 Transactions, 7 Alerts, 4 Investigators, 5 Cases	`56 Baseline Records`	**100% VERIFIED**

Production readiness & future enhancements

Honest engineering assessment of current system deliverables and roadmap.

COMPLETED DELIVERABLES

• **Rigorous 3NF Architecture:** Eliminated data redundancy and update anomalies across customer demographics, accounts, and case allocations.

• **Engine-Enforced Constraints:** Strict CHECK, UNIQUE, and FOREIGN KEY constraints reject invalid or corrupt records at the database kernel.

• **Active Procedural Surveillance:** Multi-tiered PL/SQL procedures and views automate fraud detection thresholds and case assignments.

• **Non-Repudiation Audit Defense:** Automated row-level triggers track case status transitions in forensic audit logs.

• **100% Test Pass Rate:** Fully executed and verified test suite proving both positive operational workflows and negative constraint defense.

LIMITATIONS & FUTURE SCOPE

• **Current Limitation — Synchronous Triggers:** Heavy trigger-based auditing on high-frequency transactions may impact write latency in ultra-high TPS environments.

• **Future Enhancement — Kafka Stream Ingestion:** Decouple ingest via Apache Kafka topics and micro-batched loader routines.

• **Current Limitation — Rule-Based Detection:** Surveillance rules rely on fixed thresholds (>50k INR, channel type) rather than behavioural patterns.

• **Future Enhancement — Machine Learning Inference:** Embed Oracle Machine Learning (OML) algorithms or ONNX models for real-time anomaly inference inside the database.

• **Future Enhancement — Distributed Ledger:** Implement Oracle Blockchain Tables to provide cryptographic immutability for the forensic audit trail.

FRAUDLENS • COMPREHENSIVE REVIEW 1 & REVIEW 2 COMPLETE

FraudLens: Master Project Completion

Financial Transaction & Fraud Investigation Management System

A complete, university-grade Database Management Systems implementation encompassing relational modelling, Third Normal Form schema engineering, named integrity constraints, analytical querying, PL/SQL procedural surveillance, automated case lifecycle triage, and forensic audit logging.

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